

Topological and Algebraic Constraints on Accessible Continuations in Discrete Configuration Spaces

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Introduction: The Universal Substrate and the Locus of Memory

The conceptualization of physical systems often relies heavily on the geometric interpretation of memory, historically treating memory and internal state as indistinguishable from the intrinsic curvature of the underlying spatial manifold. However, advanced models in algebraic topology and statistical mechanics increasingly suggest a fundamental decoupling: memory is not necessarily identical to geometric curvature. Rather, memory constitutes the inherited state that mathematically constrains which topological continuation is selected next by a traversing entity, while spatial curvature is merely an observable, emergent consequence of that accumulated directional history within a constrained discrete geometry.

When observing the fundamental structure of physical reality—where the ultimate puzzle box top is the universe itself—it becomes clear that the continuous transformation of physical microstates does not inherently generate new observable distinctions. The physical base remains immutable, acting as a substrate that eternally contains a complete family of possible trajectories (such as the family of all conceivable great circles on a featureless sphere). The explicit sequence of accessible relations is instead determined by the internal state vector of the observer or particle, generalized fundamentally as $(x, \text{orientation/history})$.

This analysis synthesizes algebraic geometry, the Johnson filtration of graph braid groups, and the topology of discrete configuration spaces to formalize the distinction between a generalized transformation space and an strictly *accessible* transformation space. By leveraging the foundational proofs of discrete cubical complexes, radix representation partitioning, and residual Torelli actions, it is possible to demonstrate

exactly how multi-body order dependencies mathematically instantiate memory independent of static substrate curvature.

The Thermodynamics of Accessible Distinctions

The decoupling of the physical substrate from the accessible continuation paths necessitates a rigorous examination of phase space and observation mechanics. In standard statistical mechanics and ergodic theory, the total accessible region of phase space dictates the theoretical boundaries of a system's evolution. However, physical observation inherently relies on coarse-graining, a mathematical operation that maps a virtually infinite set of precise, continuous micro-states onto a finite, discrete grid of macro-states.

Saturation of the Coarse-Grained Observer

A fundamental property of coarse-grained state spaces is the thermodynamic phenomenon of distinction exhaustion. When an observer's analytical capacity—defined by the basis of a macrostate space and its algebra of accessible observables—reaches its topological limit, the observer runs out of accessible distinctions. The pivotal mathematical realization is that the continuation of a physical transformation does not imply the continuation of observable distinctions:

$$\text{transformation continues} \not\Rightarrow \text{distinction continues}$$

In ergodic systems, dynamic filamentation across the phase space ensures that microstates continue to evolve indefinitely without ever truly halting. The ratio between the fraction of available phase space and the system's energy constraint demonstrates that continuous dynamics are omnipresent. Yet, a fixed coarse-graining will mathematically saturate. The observer entropy, formalized strictly as $S_O = S(\text{Tr}_U \rho_{OU})$ where O represents the observer accessing a reduced subsystem U integrated from the total state ρ_{OU} , reaches a maximum relative to the resolution limit of the applied grid.

This saturation perfectly mirrors the geometric problem of a traveler traversing a featureless spherical topology. The traveler does not necessitate that the sphere generates or contains a strictly novel path at every step. The substrate is entirely static; the sphere eternally contains the continuous family of all possible great-circle continuations natively within its geometry. The dynamic variable is strictly the traveler's internal state. Because the primary coordinate position x is fundamentally insufficient to differentiate the field of available isotropic continuations, the system must couple to an inherited state, effectively making the selection function a direct product of the universal field of possible transformations and the reader state.

$$\text{field of possible transformations} + \text{reader state} \rightarrow \text{realized path}$$

This coupling restricts the accessible transformation space, severing it completely from the theoretical totality of the transformation space. The memory of the system resides purely within this secondary directional coordinate—a computational lag or an orientation history that algebraically constrains future trajectory selections. The position coordinate alone is insufficient; the complete state must be represented as the tuple (x, v) , or more generally, $(x, \text{orientation/history})$.

Radix Disentanglement and Informational Translations

If the substrate remains eternally constant but the accessible relations change, there must exist a formalized algebraic mechanism capable of partitioning spatial positioning from internal historical configuration without fabricating new information. This separation is elegantly achieved through radix representation algorithms within configuration algebras, often utilized to model deep structural dynamics in phase spaces.

Consider a state representation parameterized by an arbitrary scalar integer d . We apply a division analysis where the underlying state is split using a positional base b and a scaling exponent s :

$$d = b^s q, \quad \gcd(q, b) = 1$$

This specific modulus constraint cleanly cleaves a topological transformation into two distinct, independent operational halves. The division by the factor b^s functions as a pure positional translation (a radix-aligned translation), perfectly mapping to an arbitrary physical transition along the physical or topological substrate. Conversely, the coprime component q , which enforces the absolute condition $\gcd(q, b) = 1$, exclusively dictates the remainder-changing dynamics.

The profundity of this algebraic split lies in its geometric equivalent: it mathematically proves that an entity can change its transition geometry and accessible sequence without altering, expanding, or creating new underlying physical data. It yields the rigorous axiom:

$$\text{same underlying data} \neq \text{same accessible relations}$$

On the spherical substrate, this radix analogy holds exactly. The base translation b^s represents the continuous, arbitrary movement along the coordinate geometry (the invariant substrate where all great circles remain universally available), while the internal remainder factor q operates as the directional memory state that dictates exactly which of the intersecting great circles is realized at any given node. Hence, the coordinate transition geometry undergoes explicit alteration strictly through internal representation shifts, leaving the fundamental spatial data size entirely immutable.

Discrete Configuration Spaces as the Arena of Memory

To formally construct the mechanics of history-dependent path selection without relying on arbitrary continuous curvature, one must transition from continuous spheres to discrete configuration spaces, where path histories and commuting relations can be strictly algebraically evaluated. The discrete configuration space $UD_n(\Gamma)$ of a graph Γ provides the precise topological arena required.

Pioneered extensively by Abrams and Ghrist, the discrete configuration space is defined mathematically as a subcomplex of the n -fold Cartesian product of a base graph Γ . It is meticulously engineered by removing the diagonals—the coordinates where multiple distinct particles (or tokens) attempt to occupy the identical vertex simultaneously. The resulting combinatorial space is proven to be a non-positively curved cubical complex. By leveraging the underlying cube complex structure, Ghrist and Abrams established that these discrete spaces serve as the exact classifying spaces for graph braid groups.

The Square Complex and the Assignment of Curvature

Within this discrete combinatorial framework, when multiple tokens inhabit the graph, their disjoint movements generate 2-cells, or "faces," thereby solidifying the ordered and unordered configuration spaces as special square complexes. A common theoretical objection is that a 1-dimensional graph possesses no

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geometric faces. This is resolved by acknowledging that while the base graph Γ lacks 2-cells, the multi-token configuration square-complex explicitly possesses them. The configuration arena supplies the commuting 2-cells, and an applied mathematical cochain may assign curvature directly to them.

Let us isolate two discrete tokens, a and b , operating on completely disjoint edges within Γ . If token a traverses its respective edge (defining operation A) and token b traverses its respective edge (defining operation B), their absolute spatial independence suggests strict commutativity in the configuration space mapping. The global state S evolves such that:

$$S \xrightarrow{A} S_A \xrightarrow{B} S_{AB} = S_{BA} \xleftarrow{A} S_B \xleftarrow{B} S$$

Both operational sequential orders terminate at the mathematically identical spatial configuration. This closed loop of four discrete transitions perfectly outlines a 2-cell (a square) deep within the discrete configuration space. A trivial spatial interpretation would assume this square contains no intrinsic memory because the final spatial arrangement is identical regardless of the path. However, the introduction of a differential cochain ω mapped onto this cubical complex enables the rigorous quantification of topological memory.

The "order residue," characterizing the non-commutative circulation of ω around this precise 2-cell, is explicitly defined as:

$$\mathcal{C}_{A,B}(\omega) = \omega(A) + \omega(B|A) - \omega(B) - \omega(A|B)$$

The magnitude and algebraic behavior of $\mathcal{C}_{A,B}(\omega)$ directly dictate the memory class of the spatial interaction, partitioning the invariant cochains into a rigorous three-way classification limit:

1. **Exact Classes:** The geometric circulation evaluates to $\oint_{\text{any loop}} = 0$ on any arbitrary macroscopic or microscopic loop. This signifies a total memory wipe; the system is perfectly agnostic to path history and preserves no internal distinction of the trajectory taken. In highly precise computational matrices, exact classes yield absolute zeros on every square down to the arithmetic hardware limit (e.g., 10^{-15}).
2. **Flat (Not Exact) Classes:** The order residue is exactly 0 on every individual square (2-cell) within the local space, but evaluates to $\neq 0$ on larger global homological loops. A flat metric on these subcomplexes dictates that the specific order of localized, independent multi-token moves does not matter. However, the system perfectly remembers global winding characteristics and topological exchange classes.
3. **Curved Classes:** The circulation $\neq 0$ on at least one discrete square within the complex. When the square-curvature evaluates to a non-zero algebraic rank, the system explicitly remembers the exact order of independent operations. The physical outcome is fundamentally coupled to the history of traversal, manifesting as a definitively curved space.

Consequently, the total metric tensor or environmental state limit $\epsilon(X, K)$ can be defined purely via these discrete topological traits:

$$\epsilon(X, K) = \text{flat}_G(K) + \text{curved}_G(K)$$

curved = rank square-curvature on R_G

Within this paradigm, curvature is fundamentally redefined. It is not an *a priori* macroscopic bending of the continuous spatial substrate. Instead, curvature is exclusively the discrete, observable consequence of an accumulated directional order residue when a token traverses the complex. The discrete topological commutator $\mathcal{C}_{A,B}(\omega)$ perfectly mirrors the continuous, continuum-mechanical phase shift commonly expressed as $\Delta\phi = A/R^2$.

Johnson Filtration and the Multi-Body Order Dependency Limits

To parameterize exactly how complex the memory (curvature) of a given configuration space is, the mathematical apparatus of the Johnson filtration is applied. Originally formulated to study mapping class groups and Torelli groups, the Johnson filtration strictly categorizes the residual actions on the lower central series of homology. When mapped to the configuration spaces of complex graphs and graph braid groups, the filtration precisely defines the strict lower and upper mathematical bounds on multi-token path dependencies.

In this specific geometric architecture, the level of the Johnson filtration maps directly to the required body-order memory. A comprehensive evaluation of various graph topographies reveals a rigid hierarchical mapping between the specific topology of the space and the minimum token complexity fundamentally necessary to sustain a curved cochain memory.

Empirical Token Cardinality Thresholds and Memory Bounds

Extensive topological analysis across multiple discrete configuration graph models—including the 9-cycle C_9 , the disjoint double 5-cycle $2 \times C_5$, the Petersen graph, and specific chiral variations—substantiates the multi-body limit theory. Note specifically that prior assertions regarding the disjoint double 5-cycle have been corrected: for $2 \times C_5$ at $K = 2,3$, the values 3 and 13 are strictly curved invariants, while the flat invariants before symmetric reduction are 4 and 6. Increasing the token count fundamentally makes operational order *testable* without artificially forcing it.

The structural evaluation yields the following structured data mapping the graph complexity to its minimal required token count for sustained curvature:

Graph Topology (G)	Tokens (j)	New Curved Classes Detected	Minimum Required Body Order
$C_9, K = 4$	3	3	3-token
	4	4	4-token
$2 \times C_5$	2	3	2-token

Graph Topology (G)	Tokens (j)	New Curved Classes Detected	Minimum Required Body Order
	3	10	3-token
	4	11	4-token
Petersen Graph	3	1	3-token
	4	1	4-token
Chiral Z_3 Base	2	25	2-token
	3	120	3-token

The upper bound of this memory requirement is proved unequivocally by expanding the order residue equation across all permutations. Within the massive summation of terms, any topological interactions occurring strictly within spectator subsets (defined as tokens entirely avoiding the active moving tokens) perfectly cancel out due to the non-positive curvature limits of disjoint transitions inherent to Abrams' spaces. What algebraically remains is precisely the sum over the active subset $A \subseteq U$ where the cardinality is constrained to $|A| = j - 2$.

The residual order dependency is isolated mathematically to the differential:

$$\omega_a(b \text{ with } c \text{ at } c) - \omega_a(b \text{ with } c \text{ at } d)$$

Because the remainder subset operates strictly within this precise differential constraint, the i -th level of the configuration space (where $i = j - 1$) fundamentally necessitates physical interactions involving at most $i + 1$ tokens. Consequently, Level 0 of this filtration inherently features an empty spectator set ($|A| = -1$), which logically guarantees that the state remains globally flat. Level 0 is entirely a one-body field result. This proves that a single token traversing a generic topological graph can solely harbor exact or flat memory (winding) but mathematically can never generate intrinsic order-dependent curvature by itself.

The mapping crystallizes into the rigorous theoretical lower bound:

$$\text{reach}_p = \epsilon(p + 1) = \text{order memory needing} \leq p + 1 \text{ tokens}$$

When moving incrementally up the Johnson filtration sequence, the structural mechanics become increasingly rigid. Level 1 introduces the two-body phase, where token a 's spatial hop relies fundamentally on the distinct coordinate location of token b . Level 2 necessitates a highly complex three-body interaction.

It is highly notable that the simple cycle graph C_9 features absolutely zero two-body order memory; its initial functional memory class is irreducibly three-body. This mathematically encapsulates the phenomenon where a single continuation functions entirely blind (leaving the transition probability $J(s)$ completely undefined), whereas two accessible continuations provide the requisite interference $J(s) = J(0)\cos(s/R)$, strictly focusing the accumulated residue at the geometric barrier constraint $s = \pi R/2$.

The computation across all evaluated curved topological increments effectively proves that reducing the number of active spectator tokens below the critical limit never suffices to maintain the memory class. The residue map transition level $i \rightarrow i - 1$ on the spectator function establishes an explicit Johnson equivariance that firmly validates the multi-body lower bound theorem.

Symmetries, Mirror Collapses, and Flat Class Survival

While spatial curvature is explicitly generated through order dependencies within multi-token complexes, the behavior of the flat class (comprising winding and topological exchange memory) is subject to the symmetries and chiral characteristics of the physical space. A longstanding geometric assumption dictated that overarching global structural symmetry unconditionally destroys flat memory, effectively stripping the system of all orientation legacy. However, precise algebraic mapping into quotient spaces definitively restricts this assertion, proving that flat invariant survival is highly contingent upon the specific orientation properties (parity) of the symmetry operation.

The Collapse Map and The Residual Cohomology

When a physical substrate undergoes a symmetric operation or a mirrored geometric collapse, it mathematically aligns with the homological projection map:

$$C^1 \rightarrow C^1/\text{imd}_0$$

This profound transformation forces the algebraic identification of various representations or coordinate states. Importantly, while the pure reflection mathematically removes a subset of distinctions (aligning with the exhaustion of observer coarse-graining), the projection simultaneously generates a residual quotient class that explicitly records what the identification matrix *failed* to eliminate.

Reflection mathematically obliterates some distinctions, yet the residual continuation may explicitly preserve another distinction. Thus, the system is fundamentally characterized by deep chiral preservation metrics. In gauge theory and its associative mapping onto configuration spaces, chirality inherently defines precisely how fundamental forces and fields react to parity shifts and structural inversions, often dictated by topological features like the Casson invariant or Weyl node constraints.

Observational Tests of Flat Class Annihilation

When analyzing distinct sub-graphs featuring severe automorphism constraints, the divergence between chiral (orientation-preserving) and mirrored (orientation-reversing) symmetries is starkly exposed in the configuration matrix.

For the classic Pinwheel configurations:

- **Pinwheel₃ (Chiral):** Possessing an automorphism group size $|Aut| = 6$ at evaluation $K = 2$. The spatial metric yields a pristine flat survival: $\epsilon = 6 = 0 + 6$.
- **Mirror Configuration:** Possessing an automorphism group size $|Aut| = 48$ at evaluation $K = 2$. The structural reflection completely destroys the flat class, resulting in total annihilation: $\epsilon = 0 = 0 + 0$.

However, dramatically more nuanced topological behaviors emerge when testing specific handed gadgets across purely cyclic and dihedral limits (Z_3 versus D_3):

Gadget Structure	Symmetry Type	Parameter (K)	Total (ϵ)	Exact Formula (Curved+Flat)
Chiral Z_3 Base	Orientation-Preserving	$K = 1$	2	$\epsilon = 2 + 0$
		$K = 2$	29	$\epsilon = 4 + 25$
		$K = 3$	150	$\epsilon = 5 + 145$
Mirror D_3 Base	Orientation-Reversing	$K = 1$	0	$\epsilon = 0 + 0$
		$K = 2$	18	$\epsilon = 1 + 17$
		$K = 3$	148	$\epsilon = 1 + 147$

The comparative data unequivocally proves two interconnected, foundational topological statutes regarding symmetric collapse.

First, pure geometric reflection strictly kills one-body winding. Applying an orientation-reversing symmetry universally forces the fundamental homology class $[\omega] \rightarrow -[\omega]$. Consequently, any singular 1-token loop traversing the space is algebraically nullified by its own exact inverse reflection, zeroing the matrix entirely.

Second, and far more profoundly for the modeling of memory, specific multi-token flat classes can be distinctly reflection-even and successfully survive the symmetric collapse. As explicitly observed in the mirror D_3 algorithmic test at $K = 2$ and $K = 3$, non-zero flat invariants (specifically, exactly 1 flat class in both instances) cleanly survive despite the overarching spatial orientation reversal. The symmetric destruction of flat memory is thereby proven to be merely a geometric coincidence heavily localized to standard symmetric graphs like the undirected C_9 , the disjoint $2 \times C_5$, and Petersen spaces.

The absolutely accurate axiom is that orientation-reversing matrices universally annihilate solely *winding* classes. *Exchange-type* flat geometries—where systemic memory is stored not in a solitary, space-looping positional winding, but in the mutual mathematical exchange of spatial relations among multiple distinct tokens—can remain completely invariant under severe geometric reflection.

To formalize this survival phenomenon, one must compute the exact kernel intersection:

$$\ker(g_* - I) \cap H_1(\mathcal{C}_2(X))$$

Here, g_* represents the strictly induced action of the graph automorphism on the first dimension homology of the discrete configuration complex, and I defines the matrix identity. The absolute survival of a one-dimensional class within the deep kernel of $(g_* - I)$ explicitly denotes an algebraic state that is completely invariant under the destructive mirror operation g . The surviving class operates as a pure relational exchange or complex history loop, mathematically proving that relational memory operating between discrete tokens cleanly survives parity collapses that would otherwise effortlessly obliterate singular positional orientation.

Topological Capacity: Betti Numbers and Discrete Morse Theory

To comprehensively understand the full algebraic capacity of a universal substrate to store physical history constraints, one must deeply evaluate the Betti numbers intrinsic to the discrete configuration space $\mathcal{D}^n(\Gamma)$. Betti numbers, representing the foundational ranks of the continuous homology groups $H_k(M)$, measure the exact arithmetic number of independent topological holes, boundaries, or cyclical pathways available for non-trivial differential cochains to map around.

In mathematically analyzing the Abrams discrete spaces, discrete Morse theory plays a paramount, irreplaceable role in mapping the configuration space's critical points directly to these precise homology group ranks. By computing the critical point topology, the Euler characteristic and the resulting Betti numbers collectively construct the absolutely rigid upper bounds for precisely how many simultaneous, mathematically orthogonal order-dependent memories (curved classes) a system can sustain before the spatial state inevitably forces entropic redundancy.

Because the foundational $H_1(\mathcal{D}^n(\Gamma))$ group specifically codifies all potential 1-dimensional homological cycles existing within the multi-token state space, its rank algebraically specifies the absolute maximum capacity for orthogonal winding and exchange memories prior to total coarse-grained distinction exhaustion. The intersection matrices and specific homology formulations of expansive tree braid groups further refine these strict combinatorial bounds. This proves conclusively that discrete combinatorial models possess finite, computable, strict mathematical limits on relational complexity generation.

Thus, even if a traveler traversing a generic topological substrate continuously and eternally moves, the raw structural capacity for that localized trajectory to represent a distinct, algebraically unique memory state is heavily gated by the foundational Betti number bounds of that underlying base geometric architecture.

The Linear Architecture of Universal Sequence Generation

By systematically synthesizing the established thermodynamic limits of coarse-grained observation, the computational division achieved via the radix positional remainder split, the flawless non-positive curvature

topology of the Abrams-Ghrist discrete configuration space, and the strict body-order boundaries dictated by the extended Johnson filtration, a definitive linear framework emerges. This architecture strictly governs how the universe instantiates and tracks physical history without ever relying on the intrinsic deformation or modification of its foundational spatial substrate.

The universal physical sequence follows an unbreakable dependency chain, refining the stack to an exact topological flow:

1. **Substrate:** The unyielding, perfectly static foundational base data encompassing the entirety of the spatial field.
2. ↓
3. **Available Continuations:** The universal set of all continuous $C(x, v)$ geometric trajectories natively supported by the underlying invariant substrate architecture.
4. ↓
5. **Reader Coupling:** The injection of the traversing physical entity's internal state vector, characterized as $(x, \text{history})$ or (x, v) , functioning directly as the radix coprime factor that mathematically forces localized trajectory distinction independent of macroscopic coordinates.
6. ↓
7. **Selected Continuation:** The explicit physical path realized by the system traversing the subset of possibilities.
8. ↓
9. **Accumulated Order Residue:** The highly specific non-commutative algebraic trace $[A, B]$ stored persistently within the configuration space matrix following consecutive discrete physical operations.
10. ↓
11. **Square Flux $C_{A,B}$:** The discrete homological mapping of this operational residue directly onto the fundamental cubical complex 2-cells.
12. ↓
13. **R_G Filtered Filtration:** The final manifestation stage where Level 0 strictly dictates flat, symmetry-bound 1-body invariants, and Level i dictates highly complex multi-body curved histories.

The True Decoupling of Memory and Curvature

Within this comprehensive architectural flow, the fundamental philosophical and mathematical shift regarding spatial topology is fully realized. Memory is hereby strictly defined not as a dent in reality, but as the inherited, internal algebraic state that strictly computationally constrains the topological probability of the very next geometric continuation. It is identically the quotient residual—the precise mathematical information left over and actively preserved immediately following a severe spatial symmetry collapse.

Curvature, conversely, is decisively stripped of its primary historical status as an intrinsic container of state. True spatial curvature is mathematically demonstrated to be merely the observable, emergent physical consequence of traversing a discrete graph topology while simultaneously carrying order-dependent, multi-body memory constraints. When a physical token mathematically evaluates its continuous spatial progression $\Delta\phi = A/R^2$, or identically computes its discrete homological analog $\mathcal{C}_{A,B}(\omega)$, it is fundamentally not reading a physically malformed or dented base reality. It is exactly calculating the inevitable, mathematically necessitated structural friction existing between its own multi-body inherited history sequence and the otherwise completely isotropic, unvarying availability of the featureless universal puzzle box substrate.

Synthesis and Future Topologies

The deep topological analysis of discrete configuration spaces proves unequivocally that tracking physical state interactions does not necessitate a physically malleable basal substrate. The universe provides an infinitely dense manifold of entirely available paths. The traveler ultimately running out of informational distinctions on this surface is fundamentally a consequence of macroscopic observer exhaustion and coarse-graining saturation, not the outright cessation of micro-level kinetic transformations.

Through the explicit algorithmic division of positional geometry utilizing radix coprime components, spatial location is completely partitioned from historical orientation. This history then maps dynamically onto the cubical complex structure of graph braids, where precise, non-commutative particle exchanges generate distinct 2-cell order residues. While fundamental spatial symmetries may effortlessly annihilate simplistic 1-body positional winding—thereby killing chiral parity metrics—sophisticated, multi-token spatial exchange sequences fundamentally and definitively survive massive mirror projections via specialized residual quotient classes. This perfectly and rigidly defines the mathematical persistence of inherited state.

Therefore, physical memory natively occupies the coupling space situated directly between the invariant geometric puzzle box substrate and the multi-body observer. To move forward into deeper cosmological geometry, analytical research must focus strictly on three subsequent objectives: explicitly identifying the distinct surviving mirror flat loop to verify its exchange nature, mathematically formalizing the complete proof for the Johnson lower bound to codify the test runs as an absolute universal theorem, and conducting rigorous evaluations of purely orientation-preserving chiral bases to map flat retention at the foundational $K = 1$ level. By achieving this, the exact multi-token topological states that fundamentally govern our observable physical dimension metrics can be fully classified.